

Supporting Information: Event-Timing Tests

Supporting Information

Text S1. Lohmann-style stationary occurrence test

We used a Lohmann-style moving-window event-count test to evaluate whether the complete 0.4–4 kyr KS detection-window Rousseau et al. (2023) weak and strong monsoon-start chronologies are compatible with stationary random occurrence. Each event catalogue was tested separately over the same 0–640 kyr BP analysis interval. For a catalogue with N events over duration T , the stationary rate was estimated as

$$\hat{\lambda} = \frac{N}{T}.$$

We counted the number of events $E(t_m)$ in a moving window of width $W = 20$ kyr. Window centers t_m were placed every 0.2 kyr from 10 to 630 kyr BP, so that every moving window was fully contained within the analysis interval. Under stationary occurrence, the expected number of events per moving window is

$$E_0 = \hat{\lambda}W.$$

The scalar event-count statistic was the root-mean-square departure of observed moving-window counts from this constant expectation:

$$E_S = \left[\frac{1}{M} \sum_{m=1}^M \{E(t_m) - E_0\}^2 \right]^{1/2},$$

where M is the number of window centers. Large E_S values indicate stronger temporal clustering or depletion than expected from a stationary process.

Two Monte Carlo null models were used. In the unconditional stationary Poisson null, each realization drew the total event count from $\text{Poisson}(\hat{\lambda}T)$, then assigned event ages uniformly within 0–640 kyr BP. In the conditional fixed- N null, the total number of events was fixed to the observed N , and only event ages were randomized uniformly. The unconditional null tests both total-count variability and timing variability under a stationary Poisson process, whereas the fixed- N null asks whether the observed timing pattern is unusual after conditioning on the observed event count. We generated 5000 realizations for each null model. One-sided Monte Carlo p values were computed as the fraction of simulated E_S values greater than or equal to the observed value, with a plus-one correction.

For strong monsoon starts, $N = 96$, $\hat{\lambda} = 0.150 \text{ kyr}^{-1}$, and $E_0 = 3.00$ events per 20 kyr window. The observed statistic was $E_S = 2.010$. This result is marginal under the unconditional Poisson null ($p_P = 0.083$) but significant under the fixed- N null ($p_N = 0.045$). For weak monsoon starts, $N = 100$, $\hat{\lambda} = 0.1563 \text{ kyr}^{-1}$, and $E_0 = 3.13$ events per 20 kyr window. The observed statistic was $E_S = 2.191$, significant under both the unconditional Poisson null ($p_P = 0.0266$) and the fixed- N null ($p_N = 0.0084$). These results indicate that the weak sequence is inconsistent with stationary random occurrence and that both event catalogues show non-uniform timing once the observed event count is fixed.

Text S2. Orbital phase assignment and Rayleigh tests

We used circular phase tests to ask whether weak and strong monsoon starts occur preferentially at particular precession or obliquity phases. The precession index and obliquity series were each converted into a phase variable using their local extrema. Local minima and maxima were detected, and adjacent extrema of the same type were removed so that extrema alternated. Minima were assigned phase 0, maxima were assigned phase π , and consecutive half-cycles differed by π in unwrapped phase. The unwrapped phase was linearly interpolated to event ages and then wrapped to $[0, 2\pi)$. Under this convention, precession-index minima correspond to 0° , and precession-index maxima correspond to 180° . Events whose phases required extrapolation outside the detected extrema were excluded from the corresponding Rayleigh test.

For each event type and orbital driver, we applied the Rayleigh test for non-uniformity on the circle. Given event phases θ_j , the resultant vector is

$$\mathbf{R} = \left(\sum_{j=1}^n \cos \theta_j, \sum_{j=1}^n \sin \theta_j \right),$$

and the mean resultant length is

$$\bar{R} = \frac{\|\mathbf{R}\|}{n}.$$

The mean phase is $\arg(\mathbf{R})$, and the Rayleigh statistic is

$$z = n\bar{R}^2.$$

We used the standard finite-sample correction for the Rayleigh p value. A significant result indicates departure from circular uniformity but does not by itself adjust for event history, sampling resolution, or slow background-state covariates; that adjustment is handled in the adjusted predictive Poisson model in the main text.

Strong monsoon starts show significant precession-phase clustering. Their mean precession phase is 21.9° , the mean resultant length is $\bar{R} = 0.254$, and the Rayleigh p value is $p = 0.0019$. Weak monsoon starts also show significant precession-phase clustering, but in a nearly opposite part of the precession cycle: their mean precession phase is 225.3° , $\bar{R} = 0.242$, and $p = 0.0027$. In contrast, obliquity phases do not show significant clustering. Strong monsoon starts have an obliquity-phase Rayleigh $p = 0.682$, and weak monsoon starts have $p = 0.704$. One weak-start event required phase extrapolation for obliquity and was excluded from that test, giving $n = 99$ for the weak obliquity case. These results identify precession phase, rather than obliquity phase, as the orbital phase most clearly associated with the timing of the weak and strong monsoon-start events. The obliquity Rayleigh tests and phase-ECDF diagnostics are shown in Figures S1 and S2.

Text S3. Bin-width and history-window sensitivity

The main predictive analysis used 0.2 kyr bins. We repeated the adjusted predictive-information comparison for bin widths of 0.2, 0.4, 0.6, 0.8, and 1.0 kyr. In every case, the precession-phase term remained significant after event history, sampling resolution, LR04, and CO₂. For strong starts, the likelihood-ratio p value ranged from 5.2×10^{-5} to 2.3×10^{-4} , and the preferred phase ranged from 2.1° to 5.5° . For weak starts, the corresponding p value ranged from 7.0×10^{-4} to 1.17×10^{-3} , and the preferred phase ranged from 232.8° to 234.5° . Empty-bin fractions decreased from approximately 97% at 0.2 kyr to approximately 85% at 1.0 kyr, but the qualitative phase result did not change.

We also varied the same-type event-history window used in the event-process baseline. The main text uses $W = 5$ kyr. Repeating the analysis for $W = 1, 2, 5,$ and 10 kyr preserved the precession-phase result. For strong starts, the phase-after-adjusted-climate p value ranged from 2.2×10^{-5} to 1.6×10^{-4} , with preferred phases from 2.0° to 8.1° . For weak starts, the p value ranged from 4.5×10^{-4} to 2.1×10^{-3} , with preferred phases from 227.7° to 238.4° . Thus, the phase result is not dependent on the specific 5 kyr history window. The KS-window sensitivity results are summarized in Table S1.

The main $W = 5$ kyr adjusted model ranking is shown in Figure S3. The additional-forcing sensitivity tests are summarized in Figure S4 and Table S2; predictor-correlation and fitted-rate diagnostics are shown in Figures S5 and S6.

Text S4. Composite age-uncertainty sensitivity

We tested whether plausible age uncertainty in the Cheng et al. (2016) Chinese speleothem composite could alter the event-timing results. Because a full ensemble of composite age models is not available here, we used a simple age-dependent uncertainty envelope. The assumed two-sigma age uncertainty increases exponentially from 1 kyr at 0 kyr BP to 7.4 kyr at 640 kyr BP—this envelope intentionally adopts conservative values that likely exceed the true composite age uncertainty (individual U–Th dates have two-sigma errors on the order of a few hundred years for the youngest portion). The two-sigma uncertainty was interpolated to each Rousseau et al. (2023) 0.4–4 kyr KS-window weak- and strong-monsoon start age, and the corresponding Gaussian standard deviation was taken as one half of the two-sigma range.

We generated 500 age-randomized event catalogues for each transition type. For each start age, a perturbed age was drawn from a Gaussian distribution centered on the published start age. To preserve event order, each draw was truncated by adjacent-event midpoint bounds as well as by the 0–640 kyr BP analysis interval. Thus, each randomized catalogue retained the same number and rank order of strong or weak starts as the published catalogue, while allowing individual start ages to vary within the prescribed age-uncertainty envelope.

For every randomized catalogue we repeated the Lohmann-style moving-window randomness test, the Rayleigh test for precession-phase clustering, the likelihood-ratio test for adding precession phase after the adjusted climate model, and the likelihood-ratio test comparing the full predictive model against the event-process baseline. The main orbital-phase results are robust to this perturbation. The precession-phase term after event history, sampling resolution, LR04, and CO₂ remained significant in 100% of randomized catalogues for both transition types. The full predictive model also remained significant against the event-process baseline in 100% of randomized catalogues. Rayleigh precession-phase clustering remained significant in 99.6% of strong-start and 99.0% of weak-start randomized catalogues. The weak-start Lohmann randomness result was stable under both null models, whereas the strong-start fixed- N randomness test remained significant in only 49.8% of randomized catalogues and the strong-start stationary Poisson null was never significant. The precession-phase organization is therefore not an artifact of the approximate age uncertainty considered here, but the strong-start Lohmann randomness result should be treated as a borderline diagnostic. The age-uncertainty perturbations and significance fractions are shown in Figure S7.

Text S5. Lagged predictive-information diagnostics

The main text defines the lagged predictive-information framework as a temporal-alignment diagnostic rather than a causal-delay estimator. Here we report the main diagnostic values. In the

single-driver scans after the event-process baseline, strong starts have best lags of 2.4 kyr for LR04, 1.8 kyr for CO₂, and 0 kyr for precession phase. The corresponding best information gains are 0.101, 0.120, and 0.133 bits/event. Weak starts have best lags of 0 kyr for LR04, 0.8 kyr for CO₂, and 0 kyr for precession phase, with best information gains of 0.113, 0.153, and 0.065 bits/event. The weak-start single-driver precession scan has a nominal $p = 0.013$ at its best lag but is not significant after false-discovery-rate correction ($q = 0.126$).

The conditional precession-phase scans are more relevant to the main question. For strong starts, precession phase adds information after LR04 and CO₂ at their best single-driver lags, with maximum gain at 0 kyr (0.148 bits/event, $p = 8.8 \times 10^{-5}$, $q = 3.8 \times 10^{-4}$). For weak starts, the same test also peaks at 0 kyr (0.110 bits/event, $p = 6.5 \times 10^{-4}$, $q = 0.0079$). When LR04, CO₂, and precession phase are constrained to share the same lag, the best precession-phase lag is 0 kyr for strong starts and 0.8 kyr for weak starts. These near-zero maxima indicate that event probability is most strongly organized by the contemporaneous orbital phase in this binned framework; they should not be interpreted as resolving the causal direction or response time of precessional forcing. LR04 and CO₂, in contrast, behave more like slow background controls.

The full conditional lag curves are shown in Figure S8.

Text S6. Sparse-bin and bootstrap likelihood diagnostics

The 0.2 kyr event bins are intentionally sparse. This sparsity does not by itself imply zero inflation: for an event rate of approximately 0.03 events per bin, a Poisson process predicts about 97% empty bins. We therefore compared observed empty-bin fractions with fitted Poisson expectations and computed Pearson dispersion diagnostics. In the full adjusted model, the observed and expected empty-bin fractions are 0.9701 and 0.9708 for strong starts, and 0.9691 and 0.9699 for weak starts. The corresponding Pearson dispersion estimates are 0.872 and 0.916. Thus, the main fitted models do not show evidence for excess zeros or overdispersion.

We also checked the two central nested likelihood-ratio tests using reduced-model parametric bootstrap null distributions. For each bootstrap replicate, event catalogues were simulated under the fitted reduced model, same-type event history was recomputed from the simulated catalogue, and both reduced and full models were refitted. This dynamic-history bootstrap preserves the serial dependence induced by the overlapping 5 kyr history windows, rather than treating H_i as a fixed independent covariate. We used 2000 bootstrap replicates for each event type and comparison; all refits converged. The empirical p value was computed as

$$p_{\text{boot}} = \frac{1 + \#\{\text{LR}^{(b)} \geq \text{LR}_{\text{obs}}\}}{1 + B},$$

with $B = 2000$. The bootstrap distributions are slightly heavier-tailed than the asymptotic χ^2 reference, but all four central comparisons remain significant (Table S3).

Tables

Table S1: Sensitivity of the core event-timing tests to the Rousseau et al. KS detection-window choice. Values are p-values for the complete 0.4–4 kyr and 0.6–4 kyr transition catalogues. Asterisks indicate $p < 0.05$. The “Consistent” column indicates that the same significance decision ($p < 0.05$ or $p \geq 0.05$) is reached for both KS window choices.

Event catalogue	Test	KS 0.4–4 kyr	KS 0.6–4 kyr	Consistent
Strong starts	Lohmann Poisson null	0.083	0.460	Yes
Strong starts	Lohmann fixed- N null	0.049*	0.414	No
Strong starts	Rayleigh precession phase	0.0019*	0.0010*	Yes
Strong starts	Adjusted LR04+CO ₂ after event-process baseline	0.0010*	0.0017*	Yes
Strong starts	Precession phase after adjusted LR04+CO ₂	5.16×10^{-5} *	8.76×10^{-6} *	Yes
Strong starts	LR04 after adjusted CO ₂ +phase	0.0716	0.0466*	No
Strong starts	CO ₂ after adjusted LR04+phase	0.332	0.403	Yes
Strong starts	Full model vs. event-process baseline	9.54×10^{-7} *	2.79×10^{-7} *	Yes
Weak starts	Lohmann Poisson null	0.028*	0.217	No
Weak starts	Lohmann fixed- N null	0.0082*	0.167	No
Weak starts	Rayleigh precession phase	0.0027*	0.0012*	Yes
Weak starts	Adjusted LR04+CO ₂ after event-process baseline	1.95×10^{-5} *	0.00040*	Yes
Weak starts	Precession phase after adjusted LR04+CO ₂	0.00070*	0.00031*	Yes
Weak starts	LR04 after adjusted CO ₂ +phase	0.665	0.904	Yes
Weak starts	CO ₂ after adjusted LR04+phase	0.0118*	0.0218*	Yes
Weak starts	Full model vs. event-process baseline	2.62×10^{-7} *	2.13×10^{-6} *	Yes

Table S2: Additional-forcing sensitivity tests relative to the extended baseline model. The extended baseline contains same-type event history, Cheng composite sampling resolution, LR04, CO₂, and precession phase. For “Add” tests, the reduced model is the extended baseline and the full model adds the listed forcing or forcing set. For “unique” tests, the reduced model is the model with all extra forcings except the listed forcing and the full model contains all extra forcings. ΔAICc is full minus reduced; negative values favor the added term. None of the likelihood-ratio tests is significant at $p < 0.05$.

Test	Strong ΔAICc	Strong p	Weak ΔAICc	Weak p
Add Antarctic temperature	+2.01	0.953	+1.99	0.877
Add obliquity	+1.66	0.552	+1.67	0.557
Add eccentricity	+1.87	0.712	+1.85	0.686
Add 65°N summer-solstice insolation	+1.45	0.452	-0.32	0.127
Add all four extra forcings	+7.20	0.932	+4.40	0.455
Antarctic temperature unique in full model	+2.01	0.929	+1.98	0.849
Obliquity unique in full model	+1.98	0.857	+0.82	0.274
Eccentricity unique in full model	+1.74	0.599	+1.99	0.884
65°N insolation unique in full model	+1.61	0.523	-1.20	0.073

Table S3: Reduced-model parametric bootstrap likelihood-ratio diagnostics for the two central adjusted predictive-model comparisons. The bootstrap regenerated event catalogues under the fitted reduced model, recomputed same-type event history, and refitted the reduced and full models for each replicate. This preserves the serial dependence induced by overlapping 5 kyr history windows. The empirical p value is $(1 + \#\{\text{LR}^{(b)} \geq \text{LR}_{\text{obs}}\})/(1 + B)$, with $B = 2000$.

Comparison	Event	LR	$\chi^2 p$	Bootstrap p	Exceed.
LR04+CO ₂ after event-process baseline	Strong	13.73	0.00104	0.00250	4/2000
LR04+CO ₂ after event-process baseline	Weak	21.69	1.95×10^{-5}	0.00050	0/2000
Precession phase after adjusted climate	Strong	19.74	5.16×10^{-5}	0.00050	0/2000
Precession phase after adjusted climate	Weak	14.52	0.00070	0.00200	3/2000

Figures

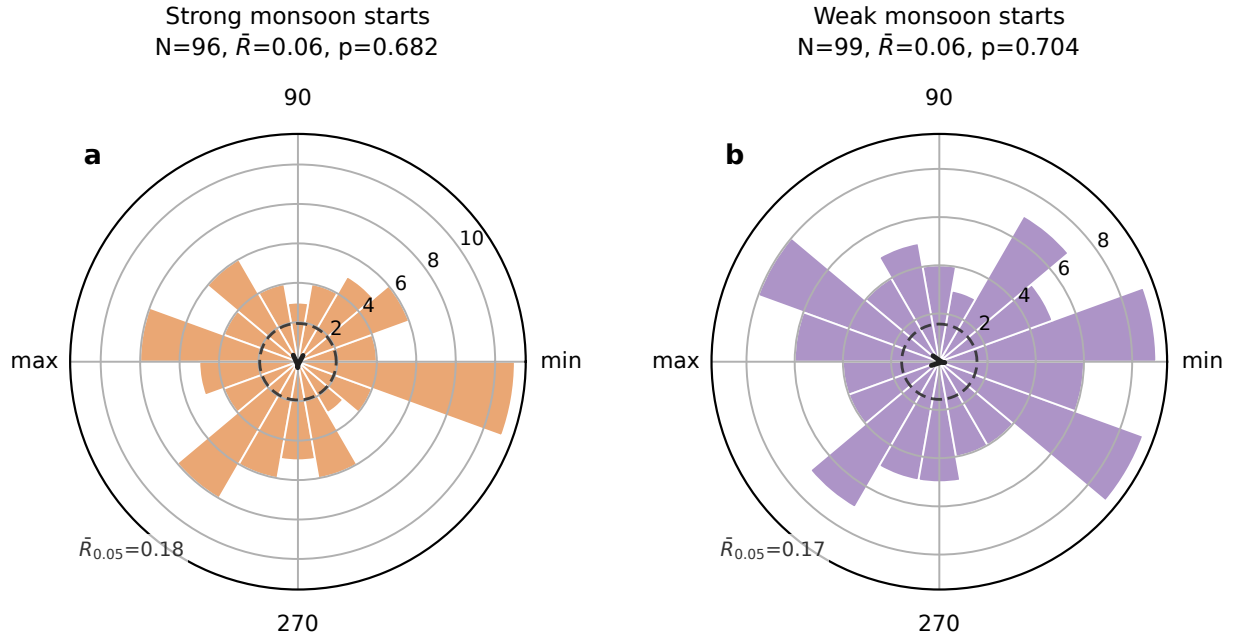


Figure S1: **Obliquity-phase Rayleigh tests.** Rayleigh tests for clustering of strong and weak monsoon starts in obliquity phase; arrows show the mean resultant direction and dashed circles mark the $p = 0.05$ threshold. Obliquity from *Laskar et al.* (2004); event catalogue from *Rousseau et al.* (2023).

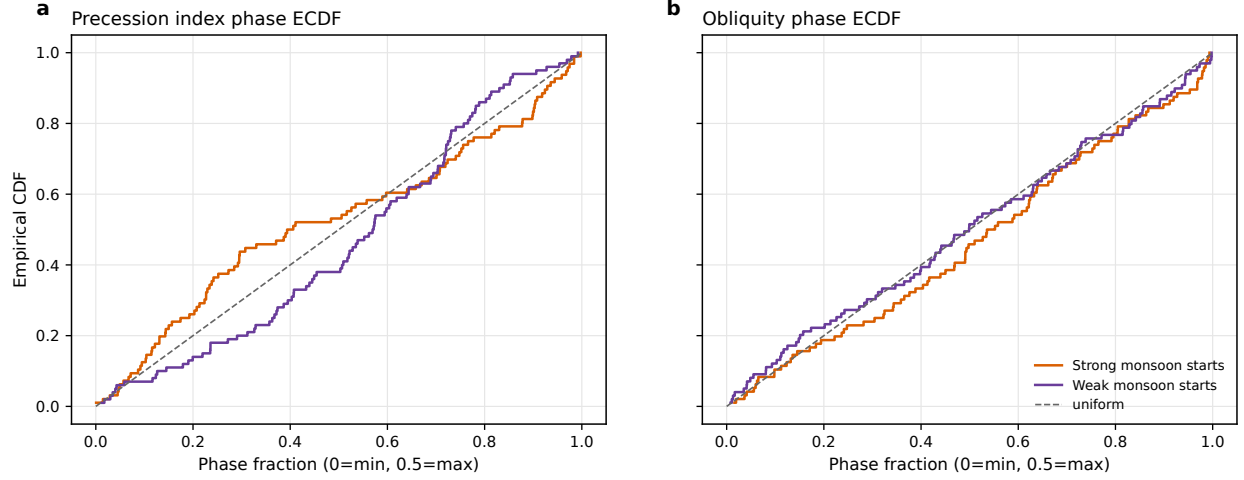


Figure S2: Empirical cumulative distributions of monsoon-start phases relative to (a) the precession index (*Laskar et al., 2004*) and (b) obliquity (*Laskar et al., 2004*). Phase fraction 0 denotes an orbital-series minimum, 0.5 denotes a maximum, and 1 denotes the next minimum. The dashed line shows the expectation for uniformly distributed phases. Curves close to the dashed line indicate no preferred phase, whereas systematic departures indicate phase clustering. Strong and weak monsoon starts show clear departures from uniformity for precession phase but remain close to uniform for obliquity.

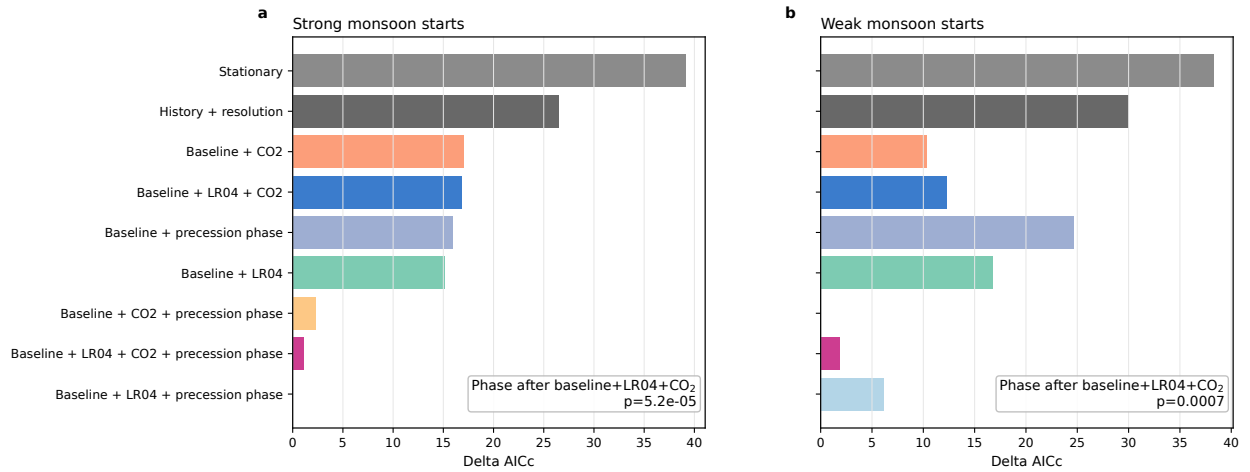


Figure S3: Adjusted predictive Poisson model comparison for (a) strong and (b) weak monsoon starts. Bars show ΔAICc relative to the lowest-AICc model within each event catalogue. The event-process baseline contains same-type event history and Cheng composite sampling resolution (*Cheng et al., 2016*); the adjusted climate model further includes LR04 (*Lisiecki and Raymo, 2005*) and CO_2 (*Bereiter et al., 2015*); and the full model adds precession phase (*Laskar et al., 2004*). Text boxes report the likelihood-ratio test for adding precession phase after event history, sampling resolution, LR04, and CO_2 .

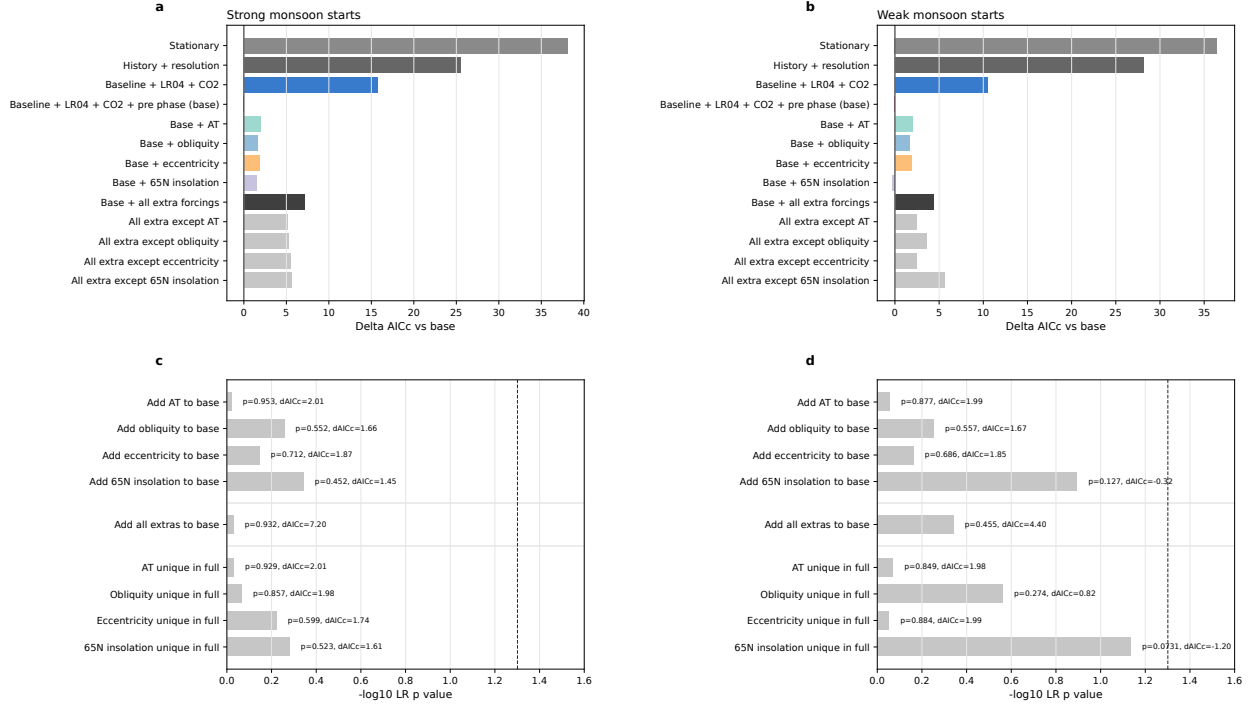


Figure S4: Sensitivity of adjusted predictive Poisson models to additional forcings. (a,b) ΔAICc for strong and weak monsoon starts, measured relative to the extended baseline containing event history, sampling resolution, LR04 (*Lisiecki and Raymo, 2005*), CO_2 (*Bereiter et al., 2015*), and precession phase (*Laskar et al., 2004*). (c,d) Nested likelihood-ratio tests for the extra forcings. Bars show $-\log_{10}(p)$, and the dashed line marks $p = 0.05$; bars to the left of this line are not significant. The three blocks test, from top to bottom, adding individual forcings to the extended baseline, adding all extra forcings jointly, and the unique contribution of each forcing within the full extended model.

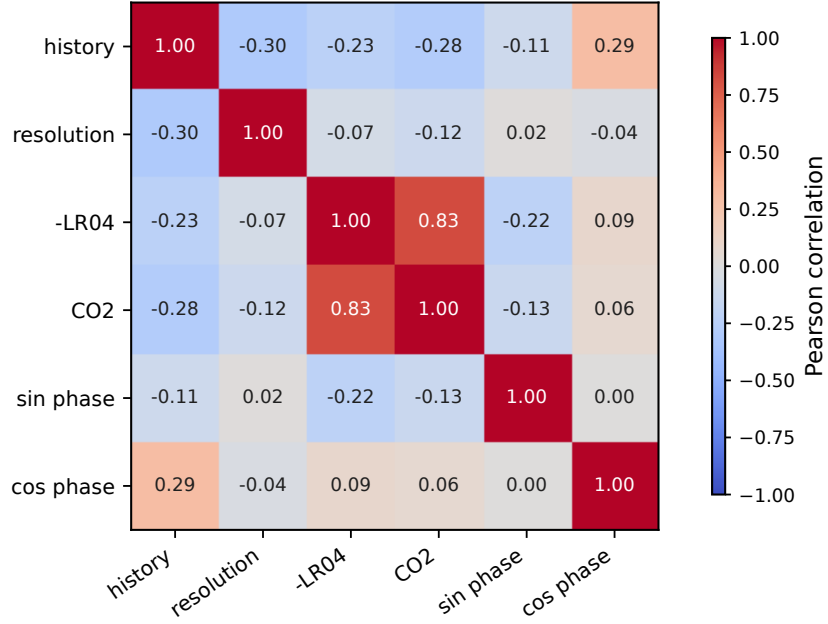


Figure S5: Correlations among predictors in the extended baseline model used for additional-forcing sensitivity tests. The matrix includes same-type event history, Cheng composite sampling resolution (*Cheng et al.*, 2016), $-LR04$ (*Lisiecki and Raymo*, 2005), CO_2 (*Bereiter et al.*, 2015), and the sine and cosine terms used to represent precession phase (*Laskar et al.*, 2004). $LR04$ is sign-inverted only for display, so its correlation with CO_2 is positive; the fitted models use the original $LR04$ orientation. The strong $-LR04$ – CO_2 correlation is consistent with treating them as partly overlapping indicators of slow glacial background state. Precession-phase terms are only weakly correlated with the event-process and background-state controls.

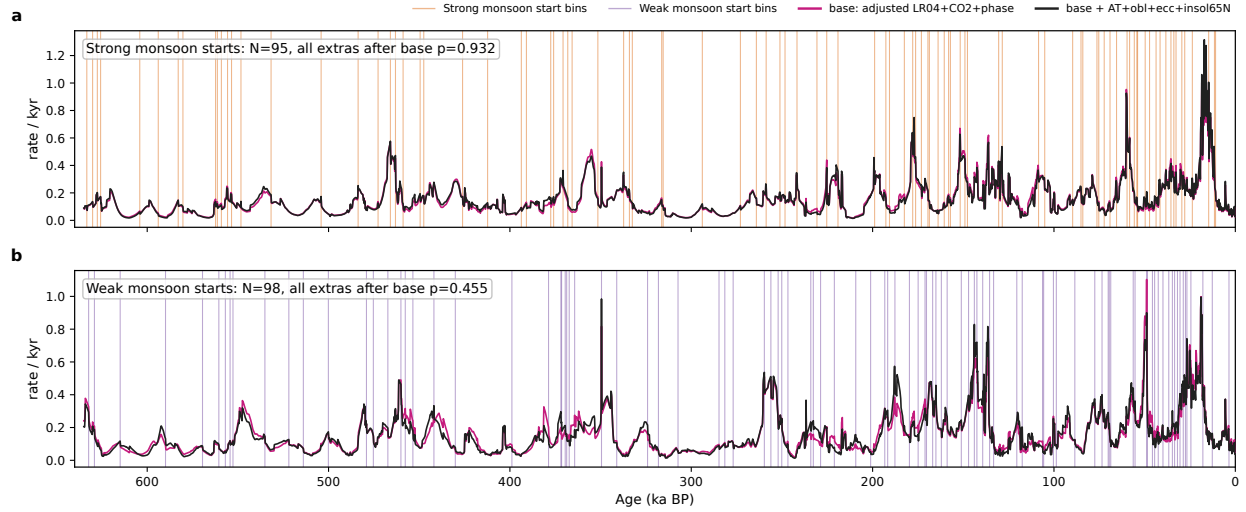


Figure S6: Fitted adjusted predictive Poisson event rates for (a) strong and (b) weak monsoon starts. Vertical lines mark observed event bins. Magenta curves show the extended baseline with event history, sampling resolution, LR04 (*Lisiecki and Raymo, 2005*), CO_2 (*Bereiter et al., 2015*), and precession phase (*Laskar et al., 2004*); black curves show the extended model after adding Antarctic temperature, obliquity, eccentricity, and 65°N summer-solstice insolation. The similar fitted rates and nonsignificant likelihood-ratio tests indicate that the extra forcings do not significantly improve the extended baseline.

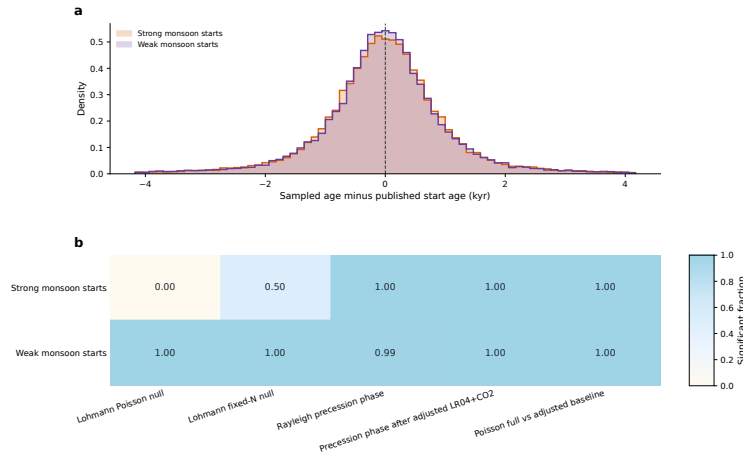


Figure S7: **Sensitivity of event-timing results to composite age uncertainty.** (a) Distribution of age perturbations applied to the published Rousseau et al. (*Rousseau et al., 2023*) 0.4–4 kyr KS-window strong and weak monsoon-start ages, after rank-preserving truncation. (b) Fraction of 500 age-randomized catalogues for which each test remains significant at $p < 0.05$. Precession-phase clustering and adjusted predictive-information results remain significant for nearly all or all randomized catalogues, whereas the strong-start Lohmann randomness result is sensitive to the null-model choice.

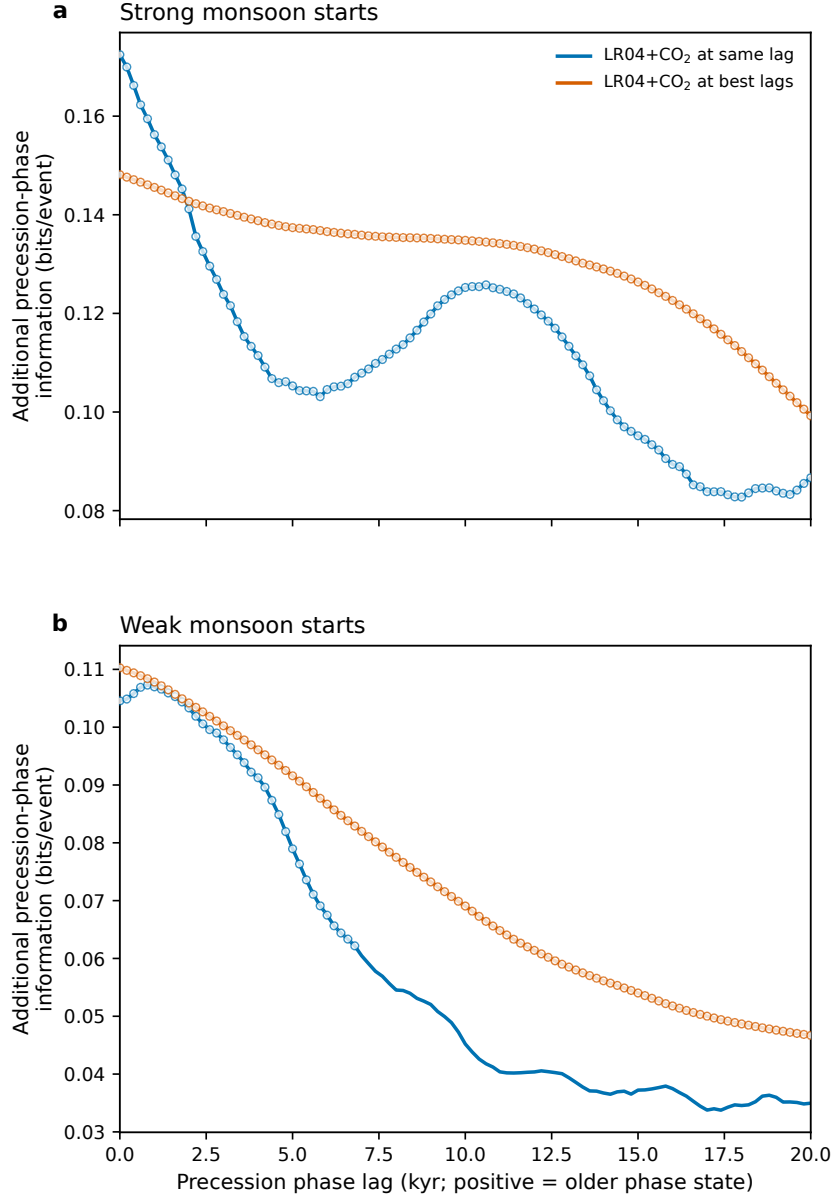


Figure S8: Conditional lagged predictive information from precession phase for (a) strong and (b) weak monsoon starts. Curves show the additional predictive-information gain, in bits per event, from adding lagged precession phase after the event-process baseline and LR04 (*Lisiecki and Raymo, 2005*)+CO₂ (*Bereiter et al., 2015*) controls. Blue curves condition on LR04 and CO₂ sampled at the same lag as precession phase; orange curves condition on LR04 and CO₂ fixed at their best single-driver lags. Open circles mark lag points that remain significant after Benjamini–Hochberg false-discovery-rate correction ($q \leq 0.05$).

References

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